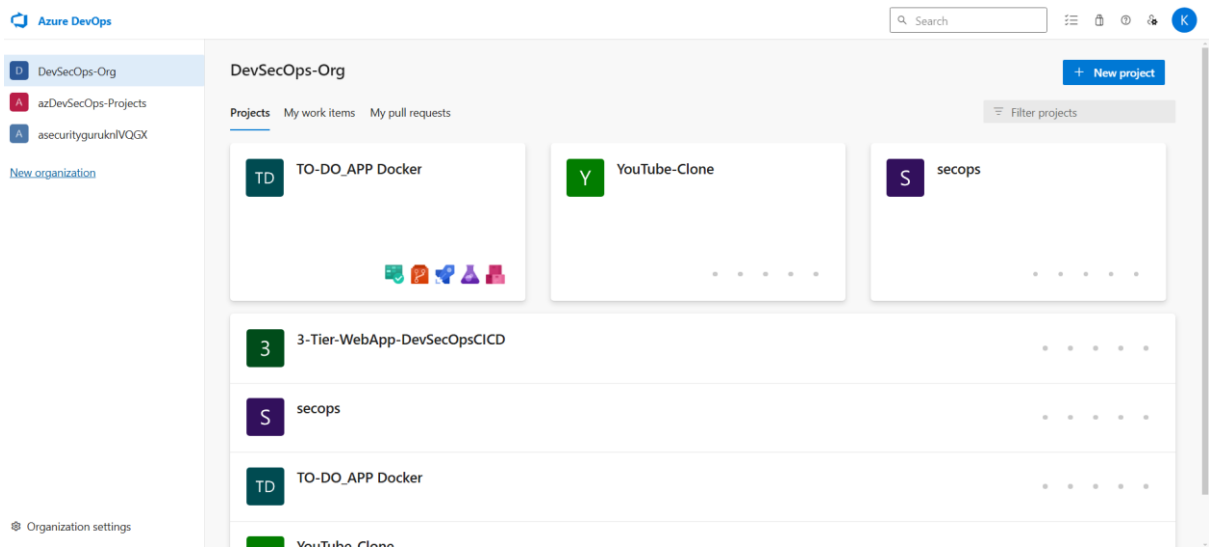
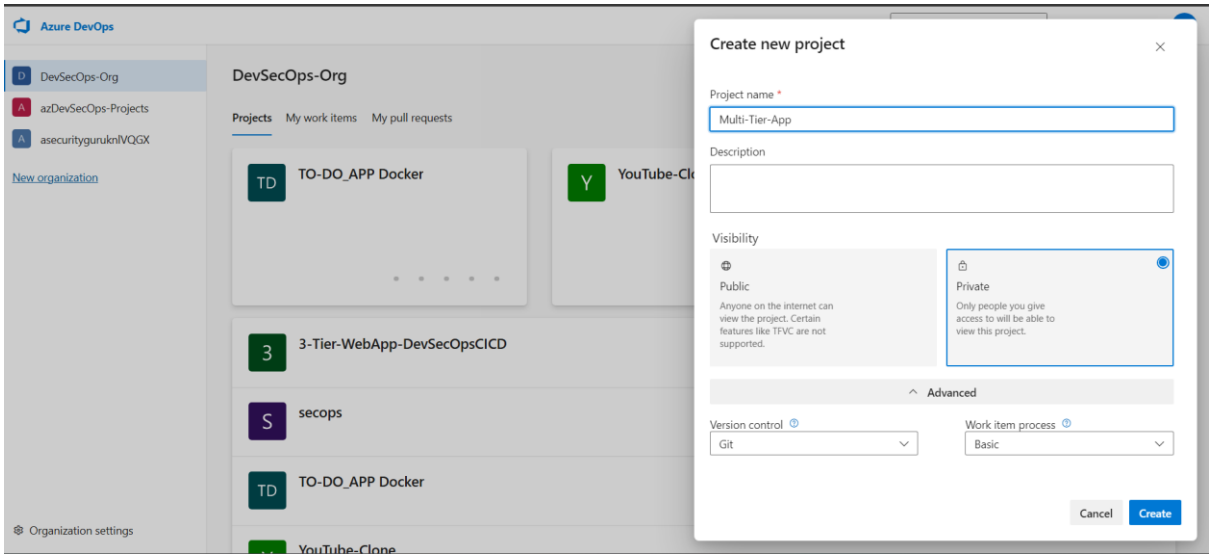
- **Deployment of Java Web App Using Azure DevOps and AKS.**

  - ***Tools and Services:***
    - **Build Tool:** Maven
    - **CI-CD Tool:** Azure Pipelines
    - **Centralized shared Repository:** Azure Repository
    - **Artifacts Repository:** Azure Artifacts
    - **Code Quality-Check & Code Analysis:** SonarQube
    - **Containerization Tool:** Docker
    - **Container Security Tool:** Trivy
    - **Container Registry:** ACR
    - **For Deployment:** AKS Cluster

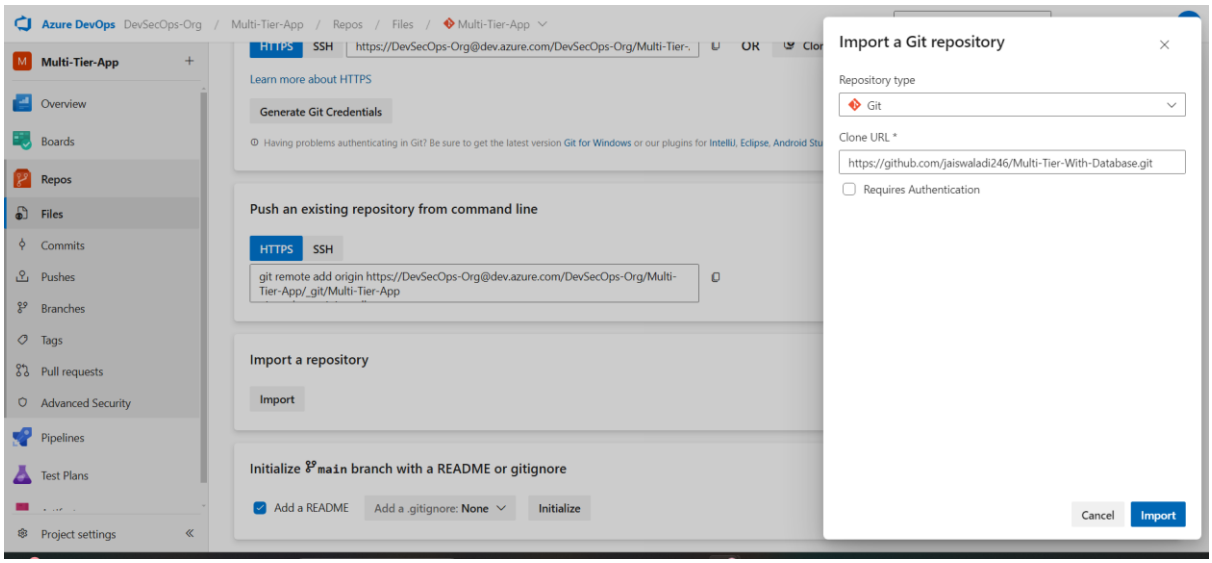
1. My Azure Organisation:



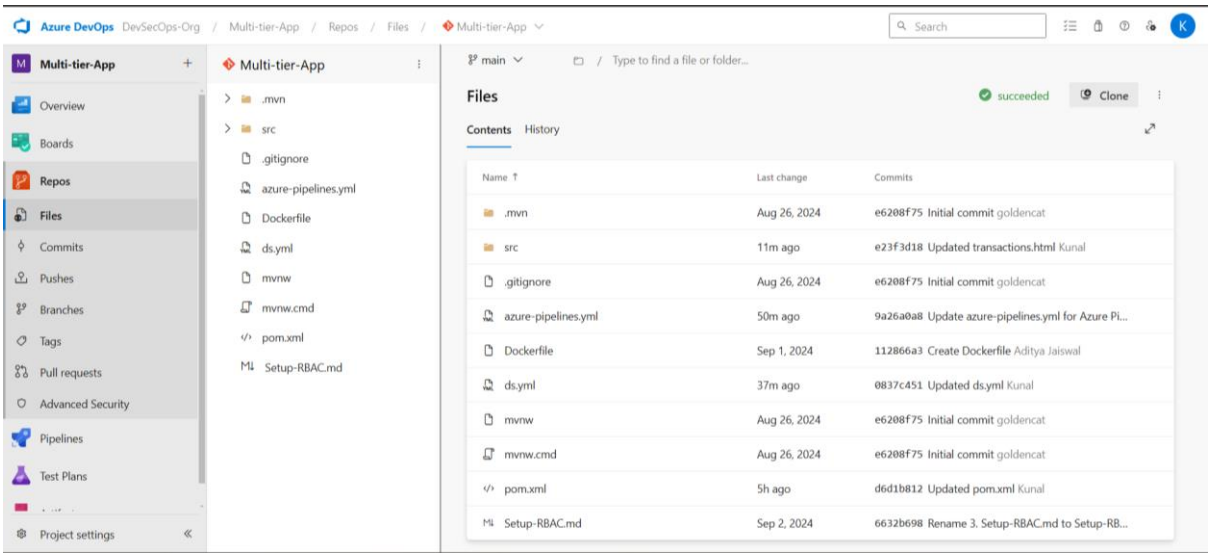
2. Create new Project:



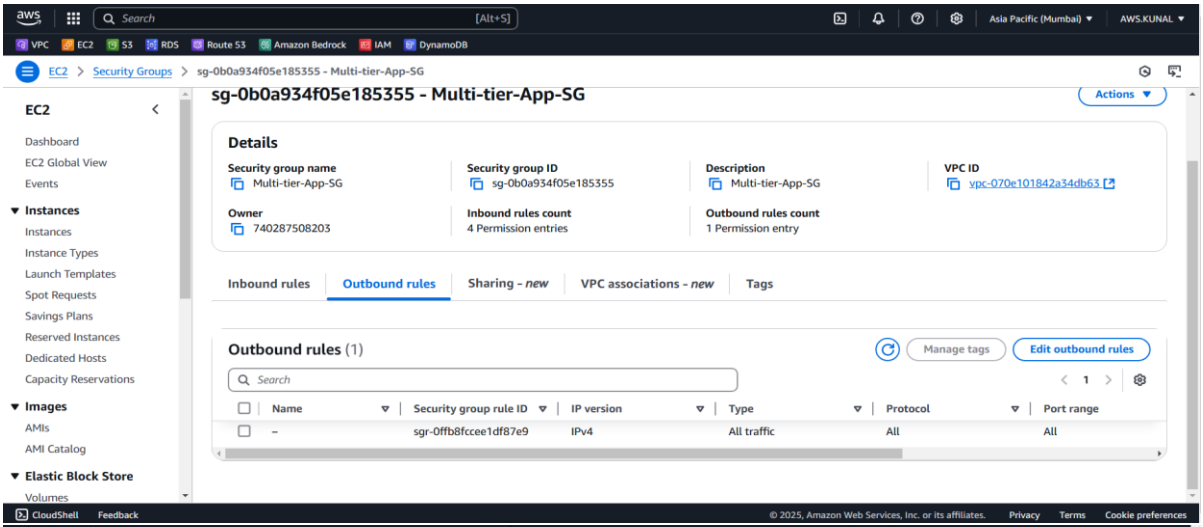
3. Import your Code repo from GitHub.

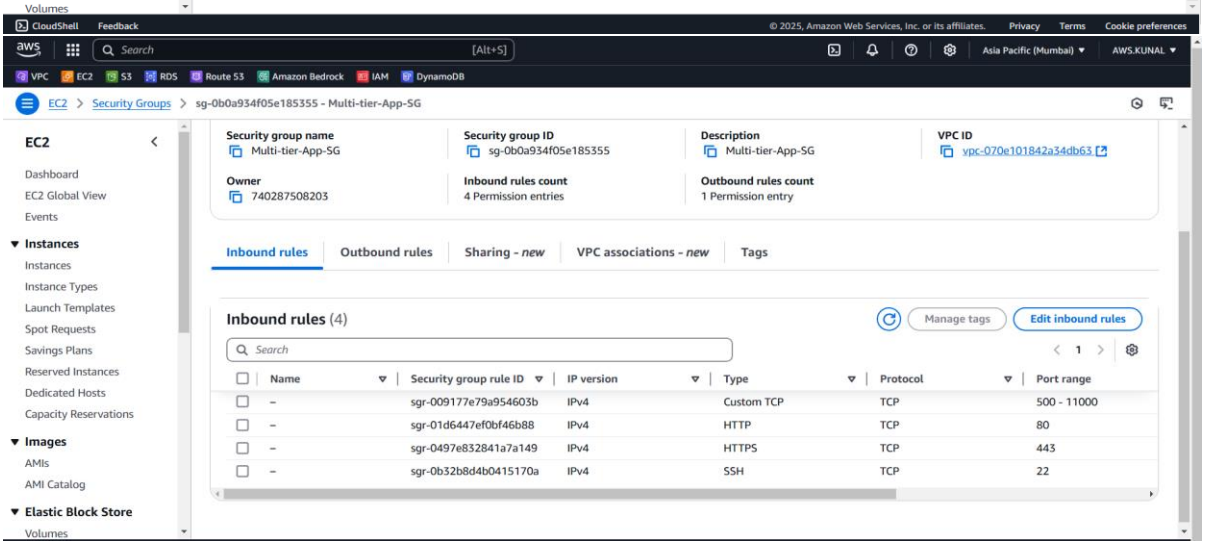


4. Repo files as below: (main branch)

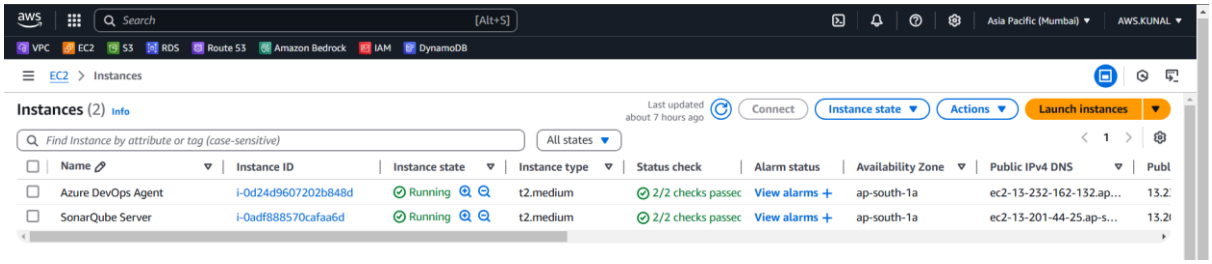


5. AWS Security Group for Azure DevOps Agent and SonarQube Server.

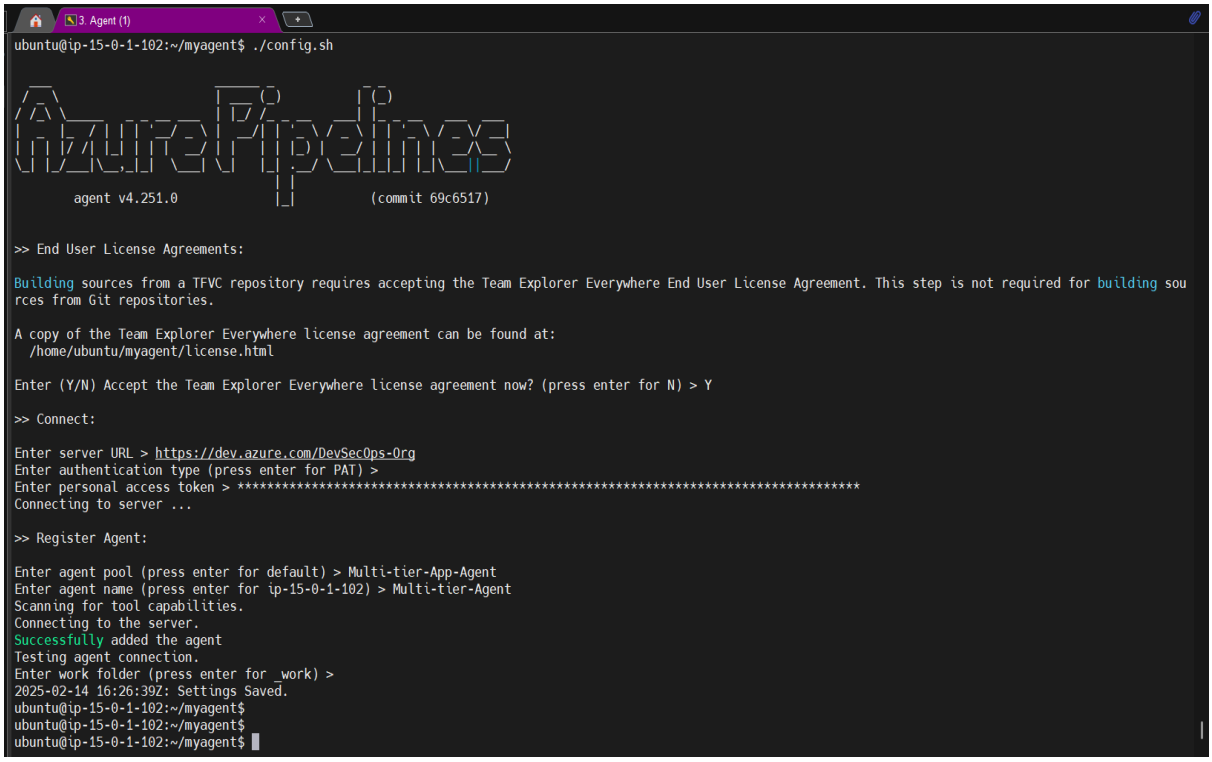




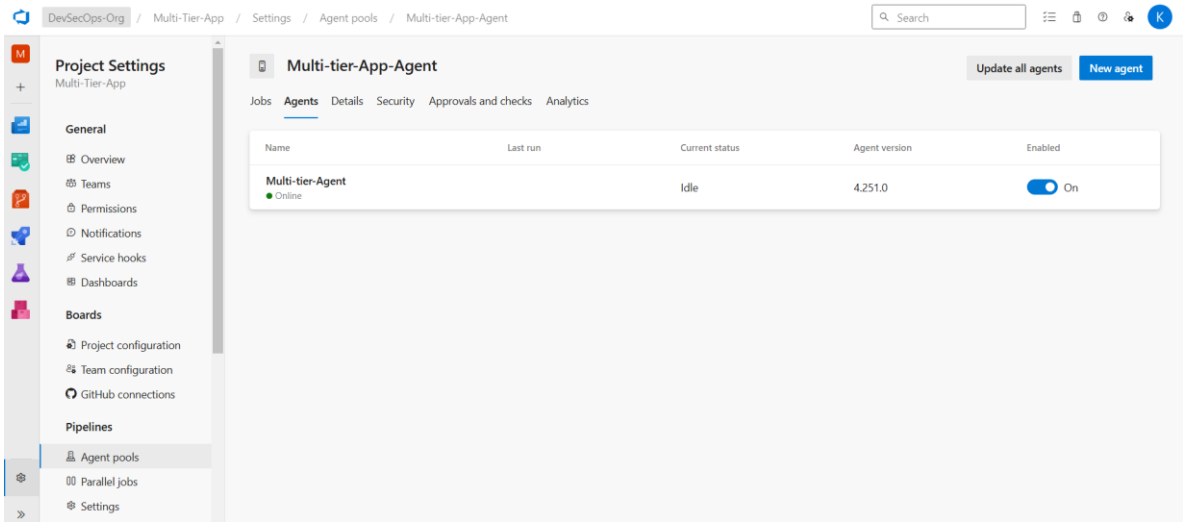
6. Azure DevOps Agent & SonarQube Server both are on AWS.



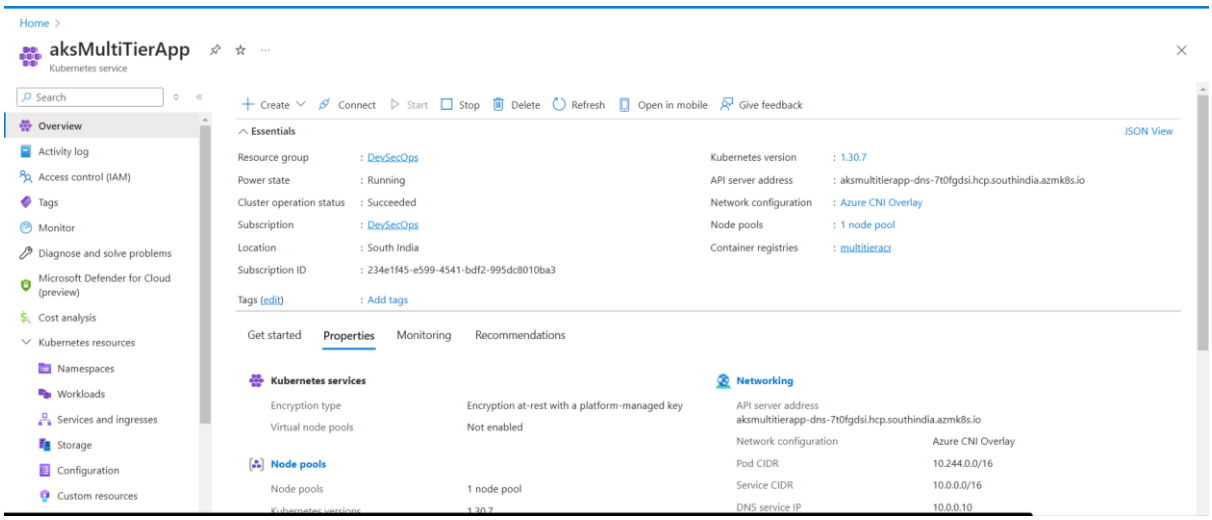
7. Configuration of Azure DevOps Agent.



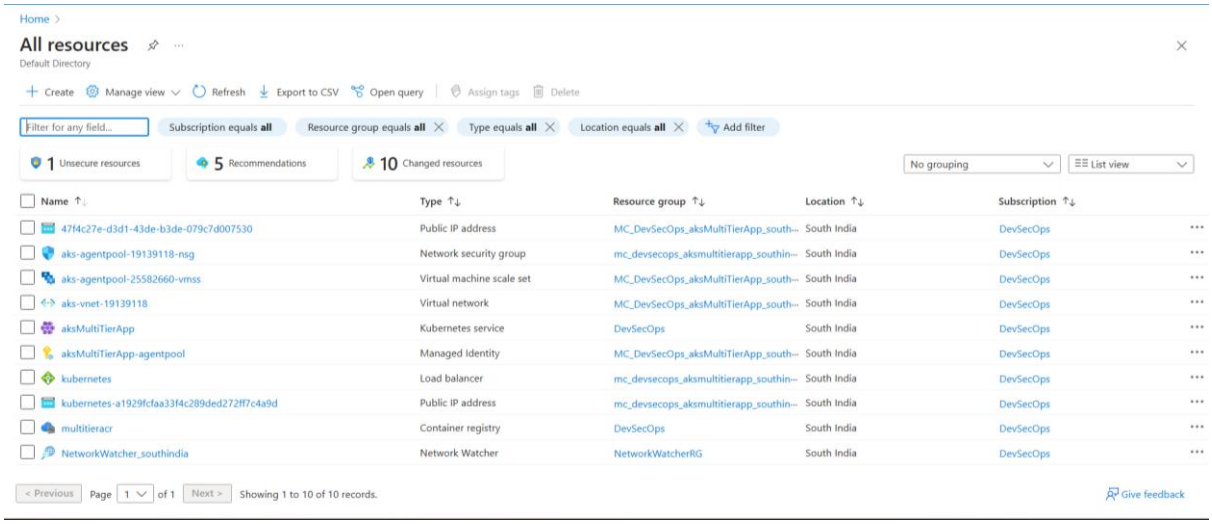
8. To make the agent Online, Run ➔ ~/myagent$ ./run.sh



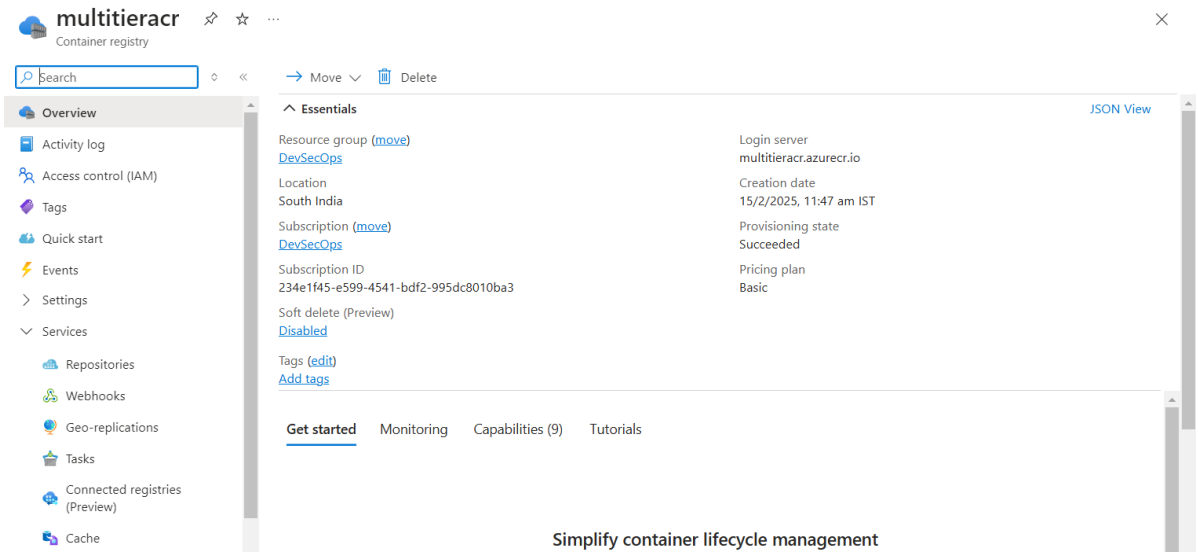
9. Creation of Kubernetes Cluster on AKS.



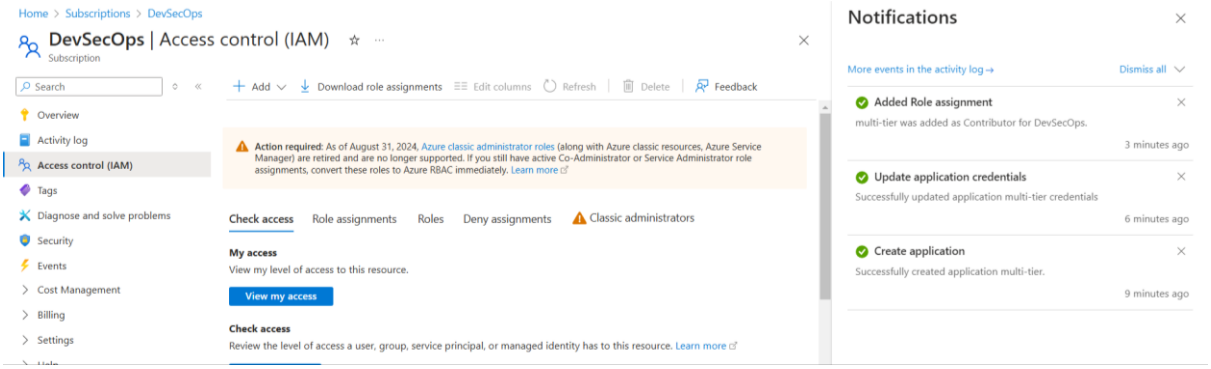
10. Created AKS resources.



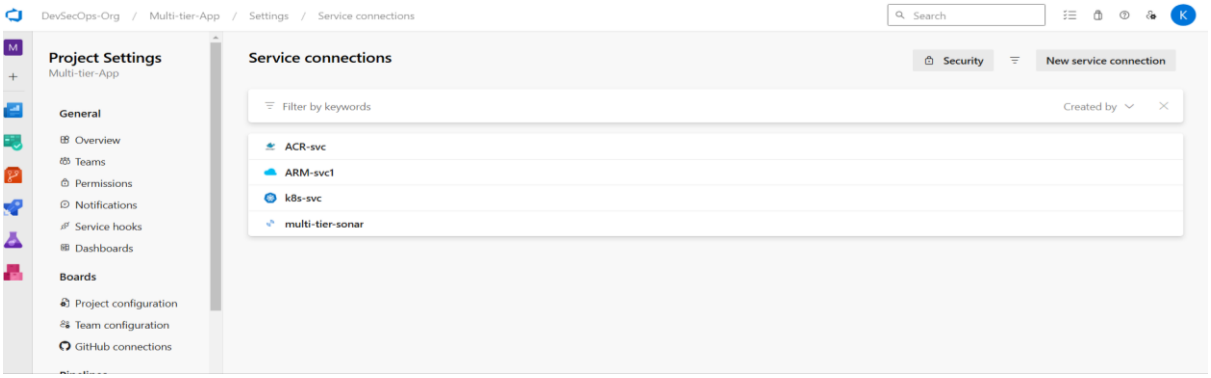
11. Creation of Azure Container Registry (ACR).



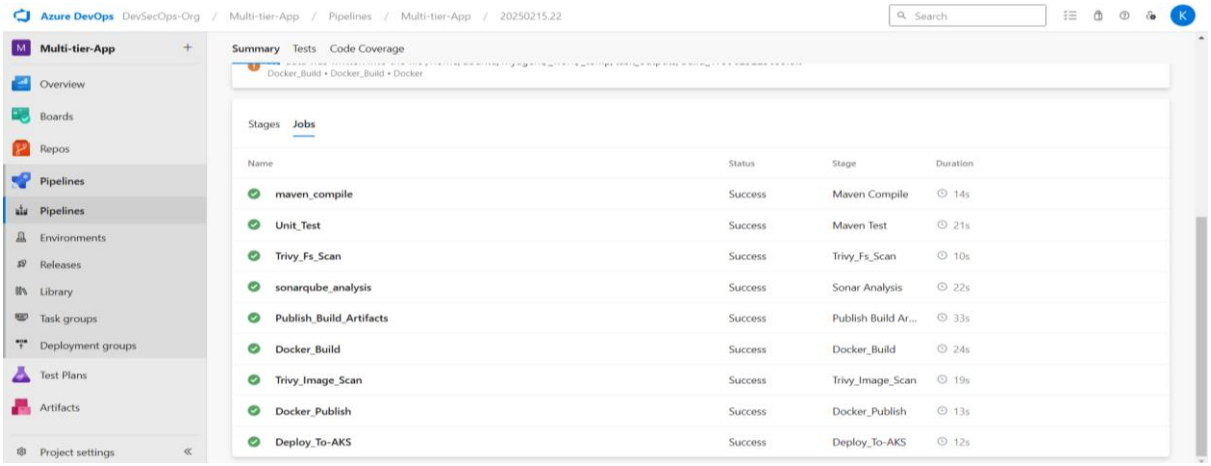
12. Created Application Service Connection ➔ Updated Application Credentials as Secret ➔ Added Role-Assignment.



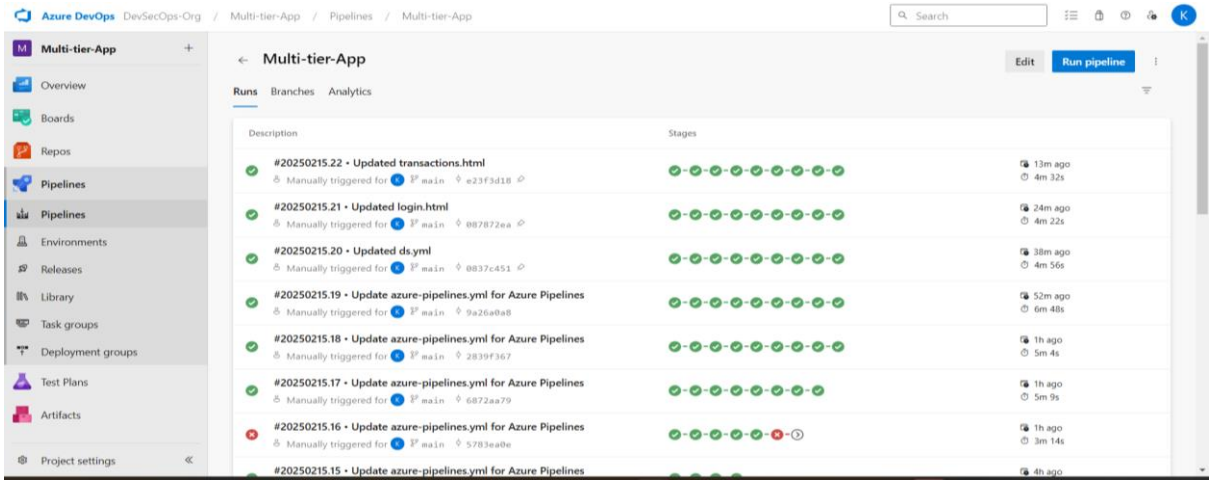
13. Below Service Connections required for ACR, K8s, and SonarQube.



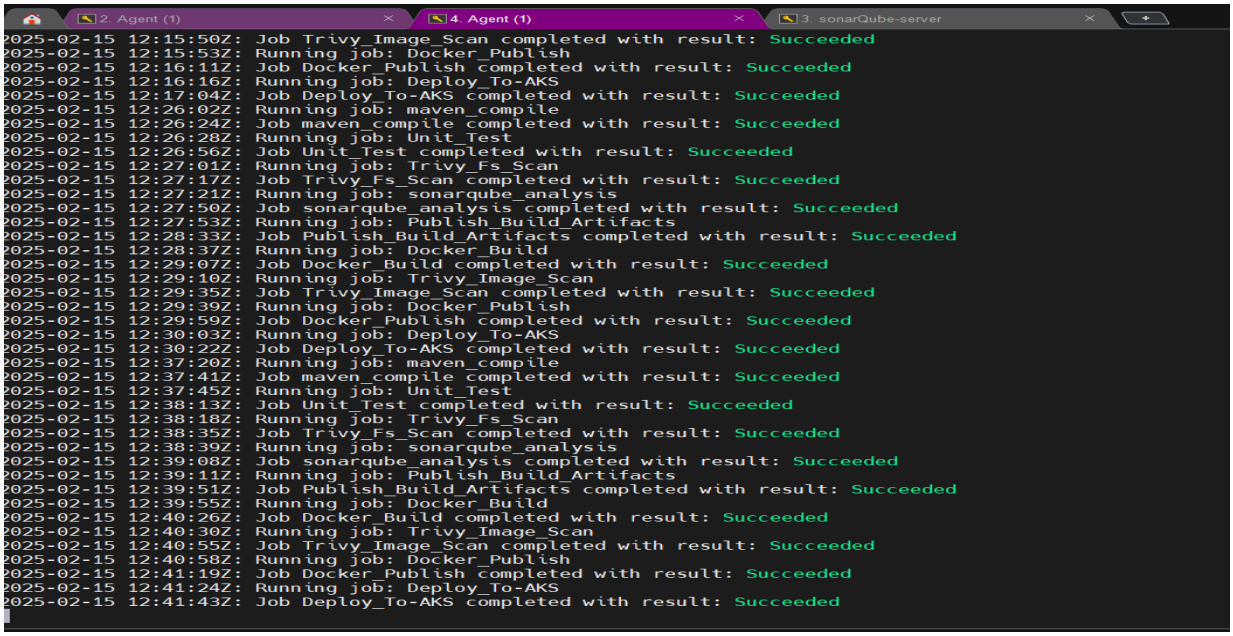
14. Azure Pipeline Stages:



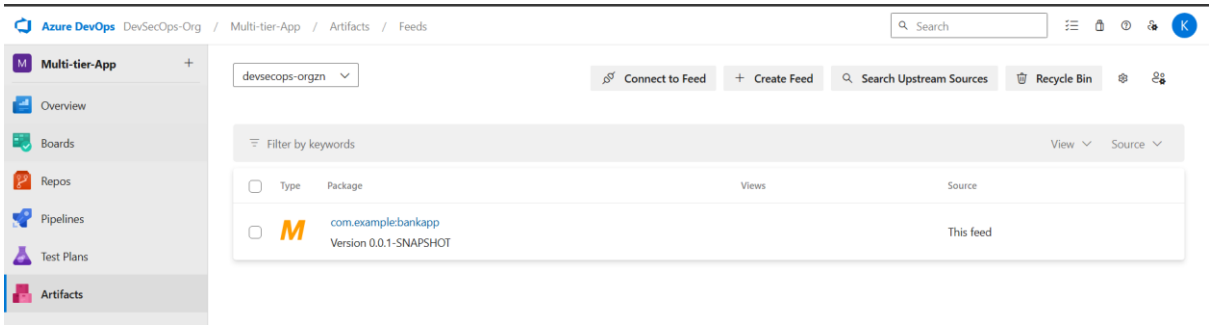
15. Finally, after a lot of debugging, pipeline is successful.



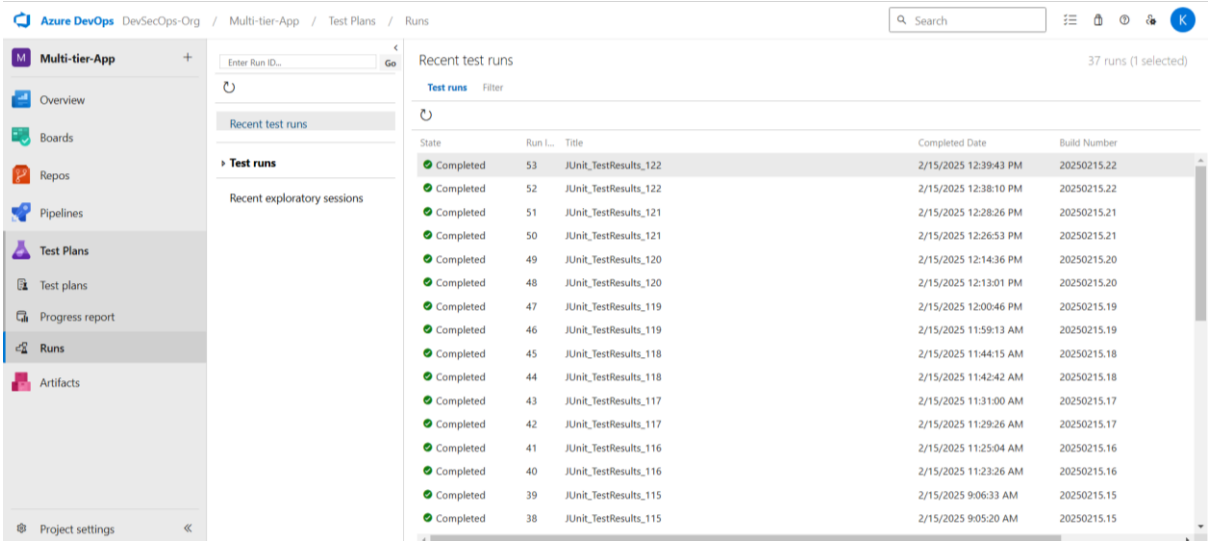
16. Pipeline status on Azure Agent.



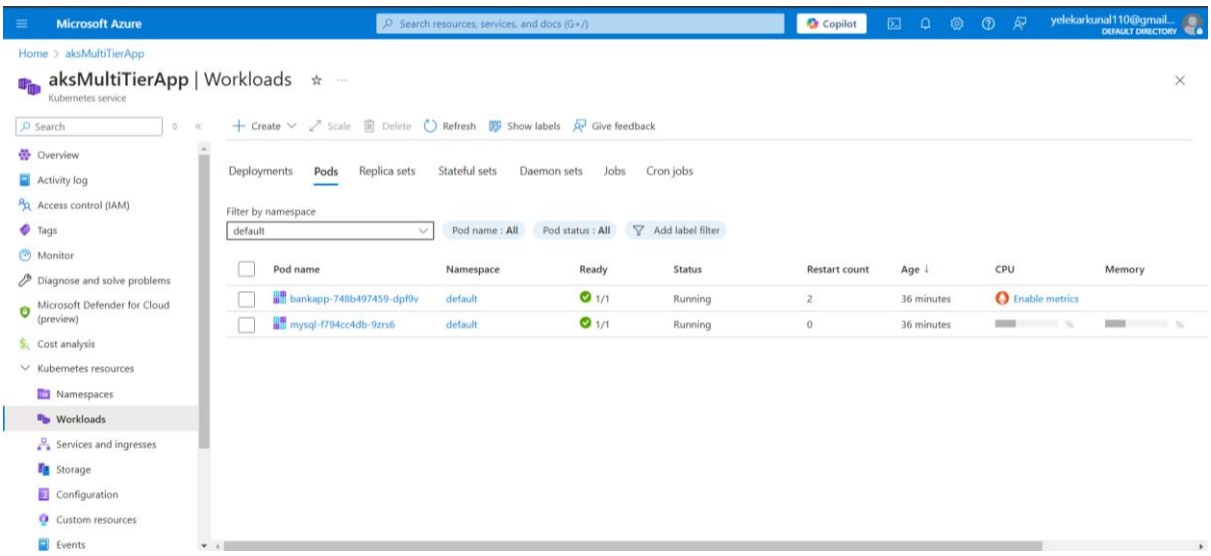
17. Build artifact (.jar) is stored in Azure Artifacts:



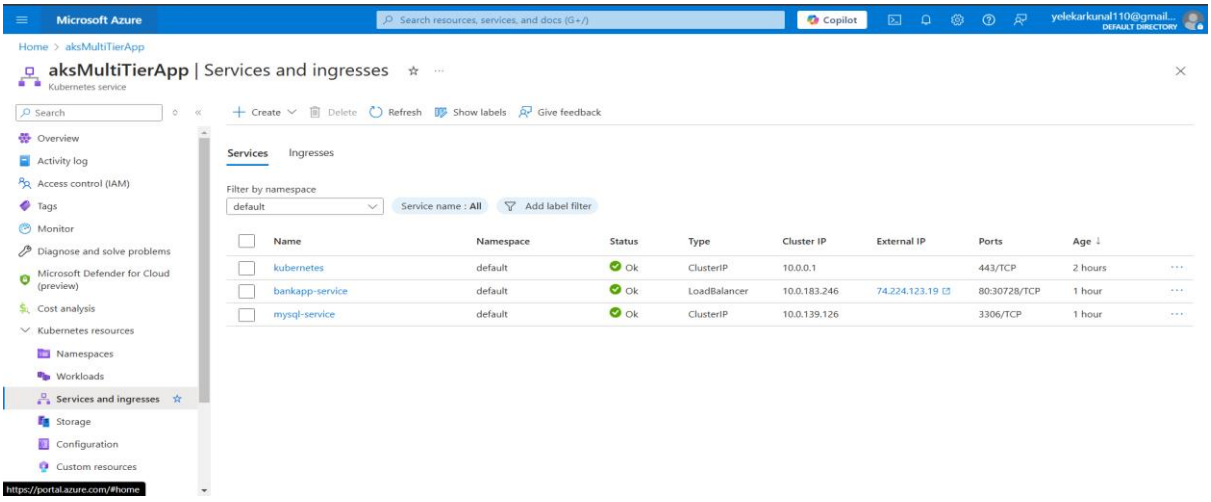
18. Test runs of the pipelines.



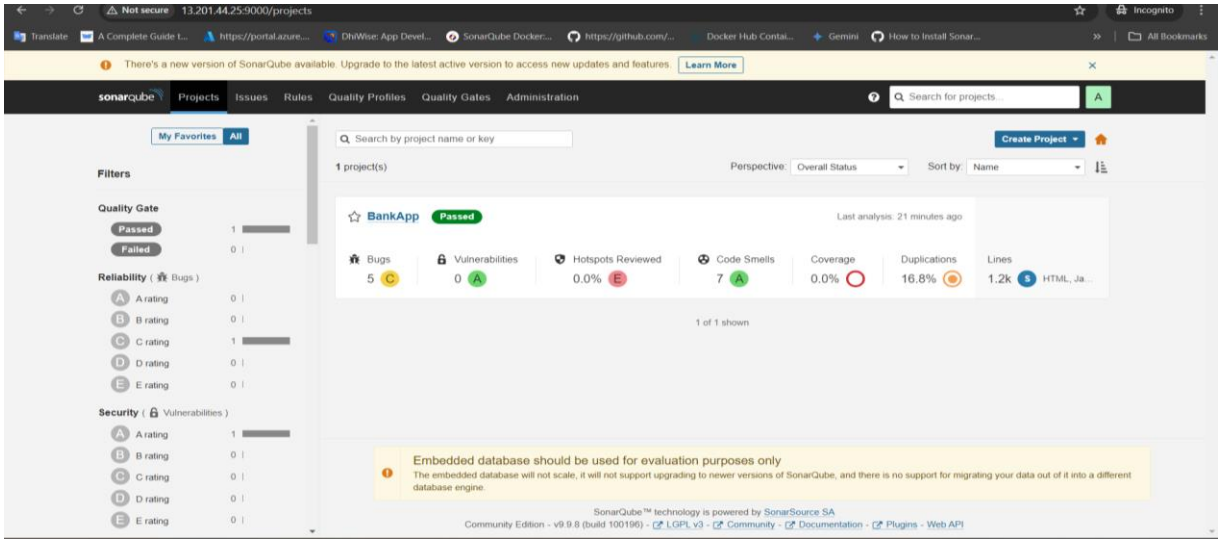
19. App is running on below 2 K8s pods.



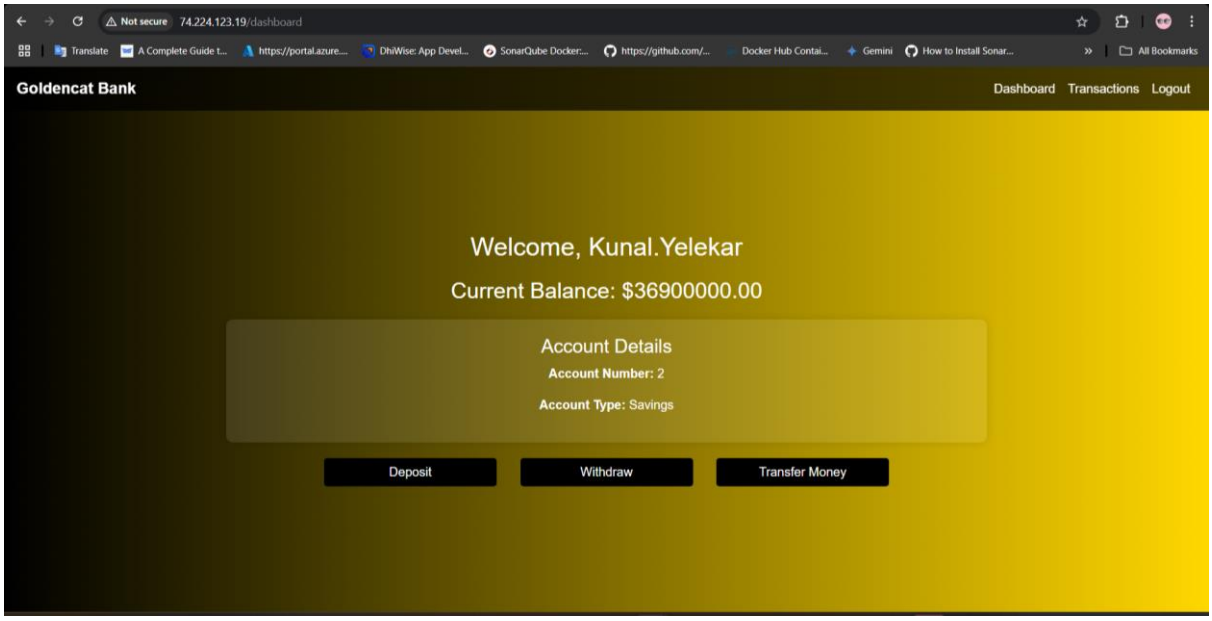
20. K8s services including Load-Balancer.



21. SonarQube Status of the Application Code Analysis.



22. Final Web Application Interface.



➔ KUNAL YELEKAR